

6-3 : Defining Data Definition Language (DDL) Practices

Overview

Assumptions

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    erDiagram
        PARENT_INFORMATION ||--o{ STUDENT : "be for"
        STUDENT ||--o{ STUDENT_ATTENDANCE : "have"
        STUDENT ||--o{ STUDENT_ATTENDANCE : "be for"
        STUDENT ||--o{ COURSE : "have"
        STUDENT ||--o{ ACADEMIC_SESSION : "given in"
        STUDENT ||--o{ EXAM_RESULT : "have"
        STUDENT ||--o{ EXAM_RESULT : "be for"
        STUDENT ||--o{ EXAM : "be for"
        STUDENT ||--o{ EXAM : "have"
        STUDENT ||--o{ EXAM : "describe"
        STUDENT ||--o{ EXAM_TYPE : "describe"
        COURSE ||--o{ EXAM_RESULT : "be for"
        COURSE ||--o{ EXAM : "be for"
        COURSE ||--o{ EXAM : "have"
        COURSE ||--o{ EXAM : "describe"
        COURSE ||--o{ EXAM_TYPE : "describe"
        FACULTY ||--o{ FACULTY_COURSE_DETAIL : "be for"
        FACULTY ||--o{ FACULTY_LOGIN_DETAIL : "be for"
        FACULTY ||--o{ EXAM_RESULT : "be for"
        FACULTY ||--o{ EXAM : "be for"
        FACULTY ||--o{ EXAM : "have"
        FACULTY ||--o{ EXAM : "describe"
        FACULTY ||--o{ EXAM_TYPE : "describe"
        DEPARTMENT ||--o{ COURSE : "offer"
        DEPARTMENT ||--o{ FACULTY : "contain"
        DEPARTMENT ||--o{ FACULTY : "belong to"
        DEPARTMENT ||--o{ EXAM_RESULT : "be for"
        DEPARTMENT ||--o{ EXAM : "be for"
        DEPARTMENT ||--o{ EXAM : "have"
        DEPARTMENT ||--o{ EXAM : "describe"
        DEPARTMENT ||--o{ EXAM_TYPE : "describe"
        ONLINE ||--o{ COURSE : "held"
        SEATED ||--o{ COURSE : "held"
        ONLINE ||--o{ FACULTY : "host"
        SEATED ||--o{ FACULTY : "host"
        ONLINE ||--o{ EXAM_RESULT : "be for"
        SEATED ||--o{ EXAM_RESULT : "be for"
        ONLINE ||--o{ EXAM : "be for"
        SEATED ||--o{ EXAM : "be for"
        ONLINE ||--o{ EXAM : "have"
        SEATED ||--o{ EXAM : "have"
        ONLINE ||--o{ EXAM : "describe"
        SEATED ||--o{ EXAM : "describe"
        ONLINE ||--o{ EXAM_TYPE : "describe"
        SEATED ||--o{ EXAM_TYPE : "describe"
        FACULTY_COURSE_DETAIL ||--o{ FACULTY : "have"
        FACULTY_COURSE_DETAIL ||--o{ COURSE : "have"
        FACULTY_COURSE_DETAIL ||--o{ EXAM_RESULT : "be for"
        FACULTY_COURSE_DETAIL ||--o{ EXAM : "be for"
        FACULTY_COURSE_DETAIL ||--o{ EXAM : "have"
        FACULTY_COURSE_DETAIL ||--o{ EXAM : "describe"
        FACULTY_COURSE_DETAIL ||--o{ EXAM_TYPE : "describe"
        FACULTY_LOGIN_DETAIL ||--o{ FACULTY : "have"
        FACULTY_LOGIN_DETAIL ||--o{ COURSE : "have"
        FACULTY_LOGIN_DETAIL ||--o{ EXAM_RESULT : "be for"
        FACULTY_LOGIN_DETAIL ||--o{ EXAM : "be for"
        FACULTY_LOGIN_DETAIL ||--o{ EXAM : "have"
        FACULTY_LOGIN_DETAIL ||--o{ EXAM : "describe"
        FACULTY_LOGIN_DETAIL ||--o{ EXAM_TYPE : "describe"
        FULL_TIME ||--o{ FACULTY : "have"
        PART_TIME ||--o{ FACULTY : "have"
        FULL_TIME ||--o{ FACULTY_COURSE_DETAIL : "be for"
        PART_TIME ||--o{ FACULTY_COURSE_DETAIL : "be for"
        FULL_TIME ||--o{ FACULTY_LOGIN_DETAIL : "be for"
        PART_TIME ||--o{ FACULTY_LOGIN_DETAIL : "be for"
        FULL_TIME ||--o{ EXAM_RESULT : "be for"
        PART_TIME ||--o{ EXAM_RESULT : "be for"
        FULL_TIME ||--o{ EXAM : "be for"
        PART_TIME ||--o{ EXAM : "be for"
        FULL_TIME ||--o{ EXAM : "have"
        PART_TIME ||--o{ EXAM : "have"
        FULL_TIME ||--o{ EXAM : "describe"
        PART_TIME ||--o{ EXAM : "describe"
        FULL_TIME ||--o{ EXAM_TYPE : "describe"
        PART_TIME ||--o{ EXAM_TYPE : "describe"
    
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The diagram illustrates the relationships between various entities in a university database. The entities and their attributes are as follows:

- PARENT INFORMATION**: # ID, * Parent 1 First Name, * Parent 1 Last Name, * Parent 2 First Name, * Parent 2 Last Name
- STUDENT**: # ID, * First Name, * Last Name, * Registration Year, * Email
- STUDENT ATTENDANCE**: * Number of Working Days, * Number of Days Off, o Eligibility for Exam
- ACADEMIC SESSION**: # ID, * Name
- COURSE**: # ID, * Name
- EXAM RESULT**: * Grade
- EXAM**: # ID, o Start Date
- EXAM TYPE**: # Type, * Name, o Description
- DEPARTMENT**: # ID, * Name, * Head
- FACULTY**: # ID, * First Name, * Last Name, * Email
- FULL TIME**: * salary, * insurance
- PART TIME**: * hourly rate
- FACULTY COURSE DETAIL**: * Contact Hours
- FACULTY LOGIN DETAIL**: # Login Date/Time
- ONLINE**: # Logon ID, * Password
- SEATED**: # Building, # Room, # Date/Time

The relationships between these entities are defined by the following connections:

- PARENT INFORMATION** to **STUDENT**: "be for" (one-to-many)
- STUDENT** to **STUDENT ATTENDANCE**: "have" (one-to-many) and "be for" (one-to-many)
- STUDENT** to **COURSE**: "have" (one-to-many)
- STUDENT** to **ACADEMIC SESSION**: "given in" (one-to-many)
- STUDENT** to **EXAM RESULT**: "have" (one-to-many) and "be for" (one-to-many)
- STUDENT** to **EXAM**: "be for" (one-to-many) and "have" (one-to-many)
- STUDENT** to **EXAM TYPE**: "describe" (one-to-many)
- COURSE** to **EXAM RESULT**: "be for" (one-to-many)
- COURSE** to **EXAM**: "be for" (one-to-many) and "have" (one-to-many)
- COURSE** to **EXAM TYPE**: "describe" (one-to-many)
- FACULTY** to **FACULTY COURSE DETAIL**: "be for" (one-to-many)
- FACULTY** to **FACULTY LOGIN DETAIL**: "be for" (one-to-many)
- FACULTY** to **EXAM RESULT**: "be for" (one-to-many)
- FACULTY** to **EXAM**: "be for" (one-to-many) and "have" (one-to-many)
- FACULTY** to **EXAM TYPE**: "describe" (one-to-many)
- DEPARTMENT** to **COURSE**: "offer" (one-to-many)
- DEPARTMENT** to **FACULTY**: "contain" (one-to-many) and "belong to" (one-to-many)
- DEPARTMENT** to **EXAM RESULT**: "be for" (one-to-many)
- DEPARTMENT** to **EXAM**: "be for" (one-to-many) and "have" (one-to-many)
- DEPARTMENT** to **EXAM TYPE**: "describe" (one-to-many)
- ONLINE** to **COURSE**: "held" (one-to-many)
- SEATED** to **COURSE**: "held" (one-to-many)
- ONLINE** to **FACULTY**: "host" (one-to-many)
- SEATED** to **FACULTY**: "host" (one-to-many)
- ONLINE** to **EXAM RESULT**: "be for" (one-to-many)
- SEATED** to **EXAM RESULT**: "be for" (one-to-many)
- ONLINE** to **EXAM**: "be for" (one-to-many)
- SEATED** to **EXAM**: "be for" (one-to-many)
- ONLINE** to **EXAM TYPE**: "describe" (one-to-many)
- SEATED** to **EXAM TYPE**: "describe" (one-to-many)
- FACULTY COURSE DETAIL** to **FACULTY**: "have" (one-to-many)
- FACULTY COURSE DETAIL** to **COURSE**: "have" (one-to-many)
- FACULTY COURSE DETAIL** to **EXAM RESULT**: "be for" (one-to-many)
- FACULTY COURSE DETAIL** to **EXAM**: "be for" (one-to-many)
- FACULTY COURSE DETAIL** to **EXAM TYPE**: "describe" (one-to-many)
- FACULTY LOGIN DETAIL** to **FACULTY**: "have" (one-to-many)
- FACULTY LOGIN DETAIL** to **COURSE**: "have" (one-to-many)
- FACULTY LOGIN DETAIL** to **EXAM RESULT**: "be for" (one-to-many)
- FACULTY LOGIN DETAIL** to **EXAM**: "be for" (one-to-many)
- FACULTY LOGIN DETAIL** to **EXAM TYPE**: "describe" (one-to-many)
- FULL TIME** to **FACULTY**: "have" (one-to-many)
- PART TIME** to **FACULTY**: "have" (one-to-many)
- FULL TIME** to **FACULTY COURSE DETAIL**: "be for" (one-to-many)
- PART TIME** to **FACULTY COURSE DETAIL**: "be for" (one-to-many)
- FULL TIME** to **FACULTY LOGIN DETAIL**: "be for" (one-to-many)
- PART TIME** to **FACULTY LOGIN DETAIL**: "be for" (one-to-many)
- FULL TIME** to **EXAM RESULT**: "be for" (one-to-many)
- PART TIME** to **EXAM RESULT**: "be for" (one-to-many)
- FULL TIME** to **EXAM**: "be for" (one-to-many)
- PART TIME** to **EXAM**: "be for" (one-to-many)
- FULL TIME** to **EXAM TYPE**: "describe" (one-to-many)
- PART TIME** to **EXAM TYPE**: "describe" (one-to-many)

Tasks

1. Create the DDL Statements for creating the tables for the Academic Database listed above – include NOT NULL constraints where necessary. (Other constraints will be added later)
2. Run/execute these commands in **Oracle Application Express**

Exercise 2: Altering the Tables

Overview

In this practice, you will:

- Alter the tables to set the constraints
- Specify a default value for a column
- Set a table to a read-only status

Assumptions

The primary and foreign key constraints are based on the ERD shown in the previous exercise and the unique constraints are based on the following :

The following fields should have unique values:

- Course Name in AD_COURSES
- Department Name in AD_DEPARTMENTS
- Student Email in AD_STUDENTS
- Faculty Email in AD_FACULTY
- Session Name in AD_ACADEMIC_SESSIONS

Tasks

1. Alter the tables in the Academic Database to define the primary key, foreign key and unique constraints.
2. Alter the table AD_FACULTY_LOGIN_DETAILS and specify a default value for the column LOGIN_DATE_TIME of SYSDATE.
3. Set the AD_PARENT_INFORMATION table to a read-only status.

NOTE: You can execute the INSERT / ALTER TABLE statements in Oracle Application Express in one of the two ways:

Method 1:

- a. Open *Oracle Application Express* and paste the commands into the SQL Commands screen one at a time and run.

Method 2:

- a. Open *Oracle Application Express* and use the same script upload method as you did with the DDL commands above.

Exercise 3: Creating Composite Primary, Foreign and Unique Keys

Overview

In this practice, you will create:

- Composite Primary Key
- Composite Foreign Key
- Composite Unique Key

**** Note – these tables are not a part of the Academic Database**

Tasks

1. Create the DEPT table with the following structure:

Column	Data Type	Description
dept_id	number(8)	Department ID
dept_name	varchar2(30)	Department Name
loc_id	number(4)	Location ID

The primary key for this table needs to be defined as a composite comprising of the dept_id and loc_id.

2. Create the SUPPLIERS and PRODUCTS table with the following structure:

SUPPLIERS TABLE

Column	Data Type	Description
sup_id	number(15)	Supplier ID part of composite primary key
sup_name	varchar2(30)	Supplier Name part of composite primary key
contact_name	number(4)	Agent Contact Name

The primary key for this table needs to be defined as a composite comprising of the sup_id and sup_name.

PRODUCTS TABLE

Column	Data Type	Description
product_id	number(10)	Product ID is the primary key
sup_id	number(15)	Supplier ID that does not hold NULL value
sup_name	varchar2(30)	Supplier Name that does not hold NULL value

The primary key for this table is product_id. The foreign key for this table needs to be defined as a composite comprising of the sup_id and sup_name.

3. Create the DEPT_SAMPLE table with the following structure:

Column	Data Type	Description
dept_id	number(8)	Department ID
dept_name	varchar2(30)	Department Name
loc_id	number(4)	Location ID

The UNIQUE key for this table needs to be defined as a composite comprising of the dept_id and dept_name.